

CO4-ATI CLASS TEST

1) if $((a < b) \text{ and } ((c < d) \text{ or } (a > d)))$ then

$Z = x * y * z$

else

$Z = Z + 1$

Sol: $L_1 = a < b$ goto L_2
goto L_5

$L_2 = c < d$ goto L_4
goto L_3

$L_3 = a > d$ goto L_4
goto L_5

$L_4 :$

$t_1 = y * z$

$t_2 = x + t_1$

$Z = t_2$

~~goto~~ goto L_6

~~$L_5 = Z + 1$~~

$L_5 :$

$t_3 = Z + 1$

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2) $(a < b) \text{ and } (c < d) \text{ or } (e < f)$

Back patching technique

$(a < b) \text{ and } (c < d)$ goto $(e < f)$

1. if $a < b$ goto L_3

if the first AND expression is false then
go $(e < f)$.

1. if $a < b$ goto 3.
2. goto 5
3. if $c < d$ goto 7
4. goto 5
5. if $e < f$ goto 7
6. goto 8
7. True
8. False

Back patching means generating jump first

3) $A = (-A + B) * (C - A + B) * C$

$$t_1 = -A$$

$$t_2 = t_1 + B$$

$$t_3 = -A$$

$$t_4 = t_3 + B$$

$$t_5 = t_4 * C$$

$$t_6 = t_5 * t_2$$

$$A = t_6$$

operators	1 st Exp	2 nd Exp	store.
Minus	A	-	t1
+	t1	B	t2
Minus	A	-	t3
+	t3	B	t4
*	t4	C	t5
*	t5	t2	t6
=	t6	-	A

A) $(a > b)$ or $(c < d)$ or $(e < f)$

L1:
if $a > b$ goto LTRUE
goto L2

L2:
if $c < d$ goto LTRUE
goto L3

L3:
if $e < f$ goto LTRUE
goto LFALSE

LTRUE:
// Boolean expression is true
goto LEND

LFALSE:
// Boolean expression is false

LEND: